

The human brain is one of the most complex biological systems, and its functional organization can be characterized as a connectivity network using functional Magnetic Resonance Imaging (fMRI)(1). In this framework, brain Regions of Interest (ROIs) are represented as nodes, while functional interactions between regions are modeled as edges. Graph theory provides a powerful framework for quantifying the topological properties of brain networks, enabling the investigation of how information is organized and exchanged across distributed neural systems(2).

A fundamental principle of brain network organization is the balance between functional segregation and functional integration. Functional segregation reflects the capacity of specialized brain regions and local networks to support distinct cognitive processes, whereas functional integration enables communication and information exchange across distributed brain systems(3). Efficient cognition requires a dynamic balance between these two complementary principles, allowing the brain to maintain specialized processing while coordinating large-scale interactions. Alterations in this balance have been associated with differences in cognitive function and neurological disorders, suggesting that network-level properties may serve as potential biomarkers of brain function(4,5). Recent evidence further indicates that higher brain-network system segregation is associated with cognitive performance, although the strength and nature of this relationship may vary across cognitive domains and populations(6).

Beyond intrinsic organization, cognitive demands can drive dynamic reconfiguration of functional networks(7). During different cognitive tasks, specific patterns of connectivity are modulated to optimize information processing and support task-specific requirements. Working-memory demands are particularly relevant because increasing cognitive load requires the coordinated reorganization of interactions within and between large-scale functional networks. However, it remains unclear whether the connectivity changes induced by increasing working-memory load carry predictive information about individual performance beyond the connectivity patterns already present under low load.

Therefore, this study examines whether functional-connectivity reconfiguration between low and high working-memory load (0-back to 2-back) predicts performance in unseen individuals using Human Connectome Project data from 336 participants. Functional connectivity was estimated for each condition across 360 cortical regions grouped into 12 networks, and reconfiguration was defined as the difference between conditions, summarized as a 78-feature connectivity fingerprint. Predictive performance was evaluated using cross-validated ridge regression against a permutation-based null. Results show that both undifferenced task connectivity ($r \approx 0.28$) and multivariate reconfiguration patterns ($r \approx 0.37$ repeated CV; $r \approx 0.28$ cross-run) predict 2-back performance, consistent with previous connectome-based predictions of working-memory performance using the same task and behavioral target but a different modeling approach (8). Although baseline connectivity contributes substantially to prediction, indicating stable individual differences, the findings reveal a pattern-versus-scalar dissociation: distributed multivariate reconfiguration patterns carry predictive information that is lost when reduced to single global measures.

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